

gemstone-rs Explorer Guide

Local HTTP explorer endpoints and safety defaults



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Explorer Guide

`gemstone-rs-explorer` is a local-only HTTP explorer built on the Rust API. It is useful for proving the browser, inspect, eval, and codegen surfaces before embedding richer tooling in VS Code or another frontend.

Install and Run

```
cargo install gemstone-rs-explorer
gemstone-rs-explorer --port 8787
```

From a checkout:

```
cargo run -p gemstone-rs-explorer -- --port 8787
```

Open:

```
http://127.0.0.1:8787/
```

The home page is a small browser UI over the same JSON endpoints. It can:

- browse dictionaries, classes, protocols, methods, and source
- run doctor/status checks
- inspect BridgeRoot and list keys
- run codegen sample/discover/preview/diff/check/generate from an editable

config path

- load and save the selected codegen config file; saving is still gated by

`--allow-write`, accepts a POST body for larger configs, and validates the config before writing

- render generated wrapper source, generated mapping config, colored unified

diff output, and a side-by-side diff in a dedicated detail pane

- remember the current browser fields locally across reloads
- put/remove BridgeRoot values with explicit string/symbol key policy and

string/small-int/bool value policy when `--allow-write` is enabled

Screenshot:

gemstone-rs Explorer

Local-only GemStone/S explorer.

- [/api/config](#)
- [/api/status](#)
- [/api/browse/dictionaries](#)
- [/api/browse/classes?dictionary=UserGlobals](#)
- [/api/browse/protocols?class=Object](#)
- [/api/browse/methods?class=Object&protocol=- all -](#)
- [/api/browse/source?class=Object](#)
- [/api/codegen/sample](#)
- [/api/codegen/preview?config=examples/codegen/gemstone-rs.codegen](#)
- [/api/codegen/diff?config=examples/codegen/gemstone-rs.codegen](#)
- [/api/codegen/check?config=examples/codegen/gemstone-rs.codegen](#)
- [/api/inspect?loop=20](#)

read_only=true allow_eval=false

Safety Defaults

The explorer:

- binds to `127.0.0.1` by default
- rejects non-loopback hosts
- starts read-only
- requires `--allow-eval` before workspace evaluation
- requires `--allow-write` before codegen writes
- reads GemStone credentials from the same `GS_*` environment as the CLI

Do not expose it publicly without adding authentication and transport security.

Browse Endpoints

Run these from a second terminal:

```
curl -s http://127.0.0.1:8787/health
curl -s http://127.0.0.1:8787/api/config
curl -s http://127.0.0.1:8787/api/doctor
```

```
curl -s 'http://127.0.0.1:8787/api/doctor?live=1'
curl -s http://127.0.0.1:8787/api/status
curl -s http://127.0.0.1:8787/api/browse/dictionaries
curl -s 'http://127.0.0.1:8787/api/browse/classes?dictionary=UserGlobals'
curl -s 'http://127.0.0.1:8787/api/browse/protocols?class=Object&meta=0'
curl -s 'http://127.0.0.1:8787/api/browse/methods?class=Object&protocol=--%20all%20--&meta=0'
curl -s 'http://127.0.0.1:8787/api/browse/source?class=Object&selector=printString&meta=0'
curl -s 'http://127.0.0.1:8787/api/inspect?oop=20'
```

Expected shapes:

```
{"status":"ok"}
{"host":"127.0.0.1","port":8787,"readOnly":true,"allowEval":false}
{"success":true,"environment":{"GS_PASSWORD":{"status":"set","masked":true}}}
{"connected":true,"sessionId":12345,"needsCommit":false,"inTransaction":false}
{"success":true,"dictionaries":["UserGlobals"]}
```

Eval

Eval is disabled by default:

```
gemstone-rs-explorer --allow-eval
```

Then:

```
curl -s 'http://127.0.0.1:8787/api/eval?source=3%20%2B%204'
```

Expected:

```
{"type":"smallInt","value":7}
```

Codegen Endpoints

The home page includes a Codegen Workflow panel. Set `Config path`, optionally enter a mapped Rust type and GemStone class, then use:

- `Load Config` to read the selected config file into the editor
- `Save Config` to POST the editor contents, validate them, and write the file

when the explorer was started with `--allow-write`

- `Sample Config` to view the starter config format
- `Discover Mapping` to ask a live stone for a mapping config proposal
- `Preview` to inspect generated Rust wrappers without writing files
- `Diff` to compare generated output with the committed file; the detail pane

shows both the exact unified diff and a side-by-side view for review

- `Check` to fail when generated output is stale
- `Generate` to write wrappers when the explorer was started with

`--allow-write`

Read-only endpoints:

```
curl -s http://127.0.0.1:8787/api/codegen/sample
curl -s 'http://127.0.0.1:8787/api/codegen/config?config=examples/codegen/gemstone-rs.codegen'
curl -s 'http://127.0.0.1:8787/api/codegen/preview?config=examples/codegen/gemstone-rs.codegen'
curl -s 'http://127.0.0.1:8787/api/codegen/diff?config=examples/codegen/gemstone-rs.codegen'
curl -s 'http://127.0.0.1:8787/api/codegen/check?config=examples/codegen/gemstone-rs.codegen'
curl -s 'http://127.0.0.1:8787/api/codegen/discover-mapping?mapped=BookingDraft&class=Object'
```

Expected check result when generated output is current:

```
{"success":true,"exists":true,"upToDate":true}
```

Write-gated endpoint:

```
gemstone-rs-explorer --allow-write
```

```
curl -s 'http://127.0.0.1:8787/api/codegen/generate?config=examples/codegen/gemstone-rs.codegen'
curl -s -X POST \
  --data-binary @examples/codegen/gemstone-rs.codegen \
  'http://127.0.0.1:8787/api/codegen/config/save?config=examples/codegen/draft.codegen'
```

Expected:

```
{"success":true,"output":"examples/codegen/generated/gemstone_wrappers.rs","bytes":1234}
{"success":true,"config":"examples/codegen/draft.codegen","bytes":512}
```

BridgeRoot Endpoints

BridgeRoot inspection and mapping-config preview use the same live session configuration as the browser endpoints:

```
curl -s http://127.0.0.1:8787/api/bridge/root
curl -s http://127.0.0.1:8787/api/bridge/keys
curl -s 'http://127.0.0.1:8787/api/bridge/get?key=BookingDraft'
curl -s 'http://127.0.0.1:8787/api/bridge/get?key=BookingDraft&key_type=Symbol'
curl -s 'http://127.0.0.1:8787/api/bridge/mapping-config?mapped=BookingDraft'
```

The browser UI exposes **Bridge key type** and **Bridge value type** controls so you can explicitly test string-key, symbol-key, string-value, small-int-value, and bool-value mappings before encoding them in a reusable config file. It persists the selected key, value, class, selector, config, and mapping fields in browser local storage; use **Clear Saved Fields** to reset the page back to the documented defaults.

Writes are disabled unless the explorer is started with **--allow-write**:

```
gemstone-rs-explorer --allow-write
curl -s 'http://127.0.0.1:8787/api/bridge/put?key=WorkbenchDraft&value=hello'
curl -s 'http://127.0.0.1:8787/api/bridge/put?
key=WorkbenchCount&value=7&value_type=SmallInt'
curl -s 'http://127.0.0.1:8787/api/bridge/put?
key=WorkbenchApproved&value=true&value_type=Bool'
curl -s 'http://127.0.0.1:8787/api/bridge/remove?key=WorkbenchDraft'
```

Expected shapes:

```
{"success":true,"name":"GemStoneRsBridgeRoot","oop":1234,"identityId":1}
{"success":true,"root":"GemStoneRsBridgeRoot","keys":
[{"printString":"BookingDraft"}]}
{"oop":5678,"classOop":9012,"printString":"aDictionary(...)" }
{"success":true,"config":"mapped = BookingDraft ..."}
{"success":true,"root":"GemStoneRsBridgeRoot","key":"WorkbenchDraft","keyType":"String",
"valueType":"String","oop":3456}
```

Product Direction

The explorer should remain a separate binary crate. The core `gemstone-rs` library stays focused on safe GemStone API access, while the explorer proves higher-level browsing and codegen workflows.

Good next steps:

- file-picker style config selection
- deeper VS Code webview integration

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